# Technical Approach



## System Architecture Flow, Technology Stack, and Open-Source Humanitarian Ecosystem

### A. TECHNICAL APPROACH & SYSTEM FLOW

**1 Tier 1 • Client & Web GIS Presentation** — APPLICATION LAYER

React 18 • TypeScript • TailwindCSS • Leaflet GIS • Vite

Interactive web GIS dashboard delivering live flood polygons, damage pins, and rescue routing.

HTTPS / REST Requests • GeoJSON Vectors & Imagery Ingestion

**2 Tier 2 • Application Gateway & Task Orchestrator** — ORCHESTRATION

Node.js (v20+) • Express.js • TypeScript • JWT Auth • Multer

Manages REST APIs, secure user sessions, GeoTIFF ingestion, and rescue squad task dispatching.

Mongoose ODM • Schema Validation & 2dsphere Spatial Queries

**3 Tier 3 • Geospatial & Document Persistence** — PERSISTENCE

MongoDB Community • Mongoose ODM • GeoJSON 2dsphere

Stores multi-temporal satellite metadata, building polygons, and mission operational logs.

Asynchronous HTTP Dispatch • Multispectral Array Streaming

**4 Tier 4 • High-Throughput AI Microservice** — AI MICROSERVICE

Python 3.10+ • FastAPI (ASGI) • Uvicorn • Pydantic • NumPy

High-speed asynchronous prediction endpoint (/predict) with automated SAR tensor normalization.

Normalized Tensors • CUDA GPU Model Forward Pass

**5 Tier 5 • Deep Learning Perception & XAI Core** — AI CORE & XAI

PyTorch Core • Segmentation Models (SMP) • Flood U-Net • Torch-CAM

Dual-stream Flood U-Net (SAR) + xBD damage assessment with Grad-CAM visual attribution maps.

### B. OPEN-SOURCE TECHNOLOGIES & USAGE RATIONALE

**PyTorch & Segmentation Models (SMP)** — DL CORE • BSD-3 / MIT

Why Used: Provides modular U-Net & ResNet architectures with native CUDA GPU acceleration, enabling full algorithmic inspectability without expensive proprietary API fees.

**FastAPI & Uvicorn ASGI Server** — AI SERVICE • MIT

Why Used: High-throughput asynchronous Python framework providing sub-second inference serving (/predict), automated request schema validation, and live OpenAPI documentation.

**React 18 & TypeScript** — FRONTEND • MIT

Why Used: Component-driven, strictly typed UI architecture delivering a resilient, zero-crash tactical dashboard for first responder incident triage and dispatch operations.

**Leaflet.js & OpenStreetMap (OSM)** — GIS MAPPING • BSD-2 / ODbL

Why Used: Zero-cost, lightweight client-side GIS library replacing expensive enterprise GIS licenses (e.g., ArcGIS) with offline tile support during disaster communication outages.

**MongoDB Community & Mongoose ODM** — DATABASE • SSPL / OPEN

Why Used: Flexible document store with native GeoJSON 2dsphere spatial indexing, enabling sub-millisecond proximity queries across hundreds of disaster-affected building footprints.

**ESA Copernicus Sentinel-1 & Sentinel-2** — EARTH OBS. • CC-BY 4.0

Why Used: Eliminates prohibitive commercial satellite imagery paywalls (5, 000–15,000 per scene), providing free, continuous, all-weather global satellite data for humanitarian aid.

#### STRATEGIC MISSION IMPACT OF OPEN-SOURCE DISASTER AI

Zero Procurement Delays | Life-Critical AI Transparency | Sovereign Offline Edge Deployment